The Supply Chain of Intelligence

A Layered Framework for Value Accrual and Defensibility in Generative-AI Software

Anand Arivukkarasu

Independent; former Product Leader, Meta (Instagram)

Correspondence: supplychainofai.com

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Abstract

This paper develops the Supply Chain of Intelligence, a descriptive framework for locating where economic value and defensibility accrue in generative-AI software. The central claim is that an AI product is not best understood as a single application or feature, but as a position across a vertical chain of layers: physical resources, infrastructure, data, models, gatekeeping, access, execution, orchestration, surface, and memory. The framework complements existing strategy vocabularies — the value chain, platform economics, the resource-based view, and Jobs-to-be-Done — by separating customer demand from supply-side durability. It introduces four stylized regularities: intelligence commoditizes downward; value accrues at bottlenecks; the surface captures attention while the chain captures power; and generation and verification should be separable in high-trust domains. It also identifies a recurring application-layer pattern, the *Defensible Triangle*, in which proprietary data, deep execution, and compounding memory reinforce one another. The contribution is taxonomic and explanatory rather than empirical: the framework is offered as a vocabulary for product leaders, founders, and investors to reason about AI-native products, not as a tested causal model or a valuation method.

Keywords: *generative AI; software strategy; value chain; defensibility; moats; platform economics; foundation models; agentic systems.*

JEL classification (suggested): *L86; O33; L10; M13; M15.*

Contents

1. Introduction: why AI needs a supply-chain map

Every wave of software produces products that look, briefly, like durable companies. The generative-AI wave has compressed that cycle. A product launches on top of a frontier model, attracts usage and attention, and may then be absorbed when a model provider, cloud platform, operating-system owner, or incumbent workflow provider ships the same capability as a bundled feature. Practitioners summarize the pattern with a crude phrase: *wrappers become features*. The phrase is useful, but it is not precise enough to guide product strategy, investment judgment, or roadmap design.

This paper offers a more precise map. Its core claim is that a generative-AI product is a **position in a supply chain**, not a point on a feature list. Value moves through that chain in stages: scarce physical resources feed compute; compute supports data and models; models pass through gates of trust, then access, execution, orchestration, surface, and memory. At each stage, a different actor may capture value. Many AI products occupy one visible layer, often the surface. Durable products either own a scarce layer deeply or combine several layers that reinforce one another.

The framework is intentionally modest in method. It is a conceptual taxonomy, not an empirical study. It does not claim that layer ownership mechanically causes survival, or that every product can be scored with precision. Its purpose is to help practitioners name structural positions otherwise discussed informally: which layer does a company actually own; which layer can absorb it; which adjacent layer could it move into; and where does durable value sit when general model capability becomes widely available?

1.1 Intended use

The framework is designed for three related decisions:

- **Product leaders** use it to identify whether a roadmap strengthens a durable layer or merely polishes an absorbable surface.
- **Founders** use it to ask which layer must be entered before the current layer commoditizes.
- **Investors** use it to distinguish a workflow with compounding advantage from a thin interface that may be bundled away.

One strategic question sits under all three: when intelligence becomes cheaper and more widely distributed, which layer stays scarce enough to support durable value? Figure 1 presents the full stack.

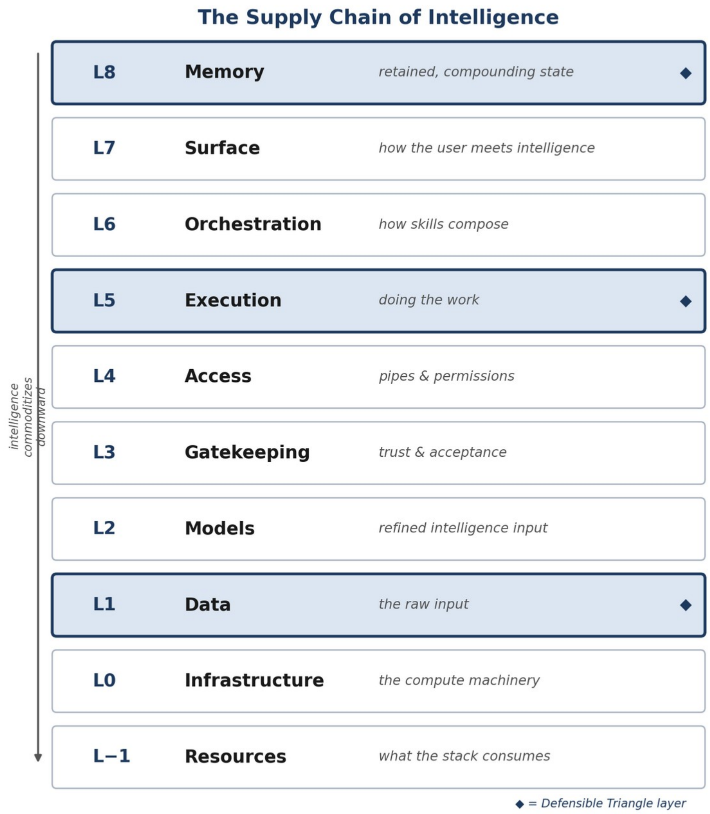

Figure 1. The Supply Chain of Intelligence — eleven layers, top to bottom. Diamonds mark the three Defensible-Triangle layers (Section 5).

2. Relationship to existing strategy frameworks

The Supply Chain of Intelligence does not replace established strategy tools. It adapts them to a software stack in which a general-purpose capability — model-based intelligence — is supplied by a small number of foundation-model providers and diffuses through APIs, copilots, operating systems, and enterprise workflows. Existing vocabularies each capture part of this shift and miss part.

2.1 Value-chain logic

Porter's value chain decomposes the firm into activities that create margin. The present framework is value-chain-like, but its unit of analysis is the industry stack rather than the internal firm; its layers are technical and organizational strata rather than business functions such as logistics or service (Porter, 1985).

2.2 Resource-based and capability views

The resource-based view asks whether a firm controls assets that are valuable, rare, hard to imitate, and organizationally embedded (Barney, 1991). The framework translates that question into stack position: proprietary outcome data, regulatory acceptance, execution playbooks, and customer-specific memory are resources tied to specific layers. This connects to the broader question of who profits from innovation when complementary assets and control points sit outside the innovator (Teece, 1986).

2.3 Platform economics

Platform economics explains why some layers capture power: they intermediate access, aggregate demand, set rules, and bundle adjacent capabilities (Rochet and Tirole, 2003; Eisenmann, Parker, and Van Alstyne, 2006). The recurring drama in AI is that a lower or broader platform layer absorbs a thinner layer above it. Platform theory explains the predator; the layered map helps identify the prey's structural options.

2.4 Jobs-to-be-Done

Jobs-to-be-Done clarifies what a customer is trying to accomplish (Christensen et al., 2016). It is strong on demand-side clarity and weak on supply-side defensibility. Two products can serve the same job and meet opposite outcomes because one owns a scarce layer while the other depends on a commoditizing capability. A compact statement of the complement: **Jobs-to-be-Done locates demand; the Supply Chain of Intelligence locates defensibility.**

2.5 AI economics and foundation models

Recent AI systems behave like broad capability substrates rather than narrow software modules. Foundation models can be adapted across tasks and distributed through APIs, embedded interfaces, and agentic workflows (Brown et al., 2020; Bommasani et al., 2021). This changes where advantage lives: the model may be necessary but is often not sufficient. Advantage instead accrues in data rights, trust gates, workflow control, evaluation, orchestration, or memory (Agrawal, Gans, and Goldfarb, 2018; Jacobides, Cennamo, and Gawer, 2018).

3. The eleven layers

The framework numbers layers from L−1 at the physical base to L8 at the top. The numbering keeps the model layer low in the stack, reflecting the argument that models increasingly behave like refined inputs to products rather than the entirety of the product. Table 1 summarizes the layers; Appendix A decomposes each into five sublayers.

Layer	Name	What it is
L8	**Memory**	What the system retains and compounds: session state, user and entity profiles, outcome histories, institutional knowledge, and learned context.
L7	**Surface**	How the user meets the intelligence: conversational, visual, embedded, transactional, and ambient interfaces.
L6	**Orchestratio n**	How skills compose into outcomes: agent loops, role routing, context and state management, human-in-the-loop design, runtime assurance.
L5	**Execution**	The work itself: domain skills, decision frameworks, retrieval-augmented workflows, operating playbooks, task-specific procedures.

Layer	Name	What it is
L4	**Access**	The pipes and permissions: APIs, protocols, access governance, real-time infrastructure, agent identity and provenance.
L3	**Gatekeeping**	Trust and acceptance: compliance, quality, safety and security, editorial review, and distribution gates.
L2	**Models**	Refined intelligence inputs: foundation, specialized, embedding and retrieval, routing, and reasoning models.
L1	**Data**	The raw and structured input: public, proprietary, behavioral, outcome, and synthetic data; licensed corpora.
L0	**Infrastructure**	Compute machinery: silicon, data centers, interconnect fabric, cloud compute, edge and on-device compute.
L−1	**Resources**	What the stack consumes: energy, thermal and water management, fabrication capacity, critical materials, skilled trades.

Table 1. The eleven layers of the generative-AI stack, top to bottom.

Two design choices are central. First, **memory sits at the top.** As model capability diffuses, the durable asset often migrates toward what the system accumulates and cannot re-derive from a public model — customer state, outcome histories, institutional preferences, operational context. Second, **gatekeeping is a first-class layer.** Where outputs carry regulatory, fiduciary, safety, or reputational weight, the right to be trusted is itself a scarce position.

4. Four structural regularities

Four regularities follow from the layered view. They are called *laws* for mnemonic clarity, but should be read as **stylized mechanisms rather than scientific laws.** Each can be resisted by distribution, capital, timing, regulation, or exceptional execution.

Law I — Intelligence commoditizes downward

A product whose only differentiator is general model capability is exposed to absorption by the layer below it. When the claim is only that the model can perform a generic task, and the provider can ship that task broadly, the product's distinct value narrows. Distribution and switching costs delay this; they do not remove the underlying pressure.

Law II — Value accrues at bottlenecks

Durable margin tends to sit where the chain is scarce and hard to replicate: proprietary data, regulatory acceptance, distribution control, domain execution, compounding memory. The strategic question is not only what the product does, but which scarce layer it controls and whether that scarcity will persist.

Law III — The surface captures attention; the chain captures power

A polished interface wins usage, narrative, and early growth, yet a surface unconnected to a deeper control point is vulnerable. Enduring products pair a usable surface with ownership of another layer — data, gatekeeping, execution, orchestration, access, or memory. Attention can be rented; structural power must be owned or contractually protected.

Law IV — Generation and verification should be separable

Where outputs carry fiduciary, regulatory, safety, or reputational weight, the actor that generates a result and the actor that verifies it often need to be separable; a generator certifying its own high-stakes output creates a trust problem. This is why gatekeeping can become an independent value-bearing layer rather than a feature of the generator. The law is partly descriptive and partly normative — it describes where value may accrue and where trust should sit if the system is to be credible.

5. The Defensible Triangle

For application-layer firms that do not own compute or foundation models, a recurring durable pattern combines three reinforcing layers (Figure 2):

1. **Proprietary data (L1)** — inputs, rights, or observed behavior the model layer cannot easily acquire.
2. **Deep execution (L5)** — domain skills, decision procedures, and operating playbooks encoded into the workflow.
3. **Compounding memory (L8)** — customer-specific or network-level state that improves with use and raises the cost of leaving.

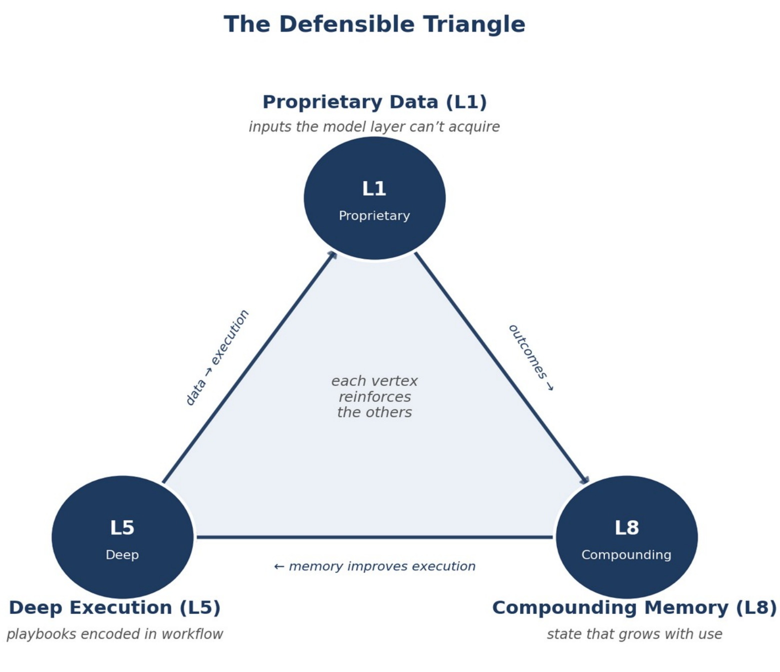

Figure 2. The Defensible Triangle. Data improves execution; execution generates outcome data; outcomes compound into memory; memory improves future execution.

A competitor may replicate a visible feature, but cannot instantly reproduce a customer's accumulated state, the workflow-specific judgments embedded in execution, or the historical feedback loop connecting them. The triangle is not the only route to defensibility: a firm may also survive by owning one scarce layer with unusual depth — infrastructure control, licensed data, regulatory access, or a trusted distribution gate. The triangle is a pattern for application-layer firms, not a universal rule.

6. Illustrative applications

The cases below are **illustrative archetypes, not a validated sample.** They were selected because they clarify the categories, and the trajectory descriptions are consistent with public reporting as of early 2026 rather than independently audited. Each links to the longer case note in the author's public corpus. A later empirical version should replace these with coded cases, explicit data sources, and falsifiable predictions (Section 7.5). The point here is to show how the vocabulary changes the diagnosis.

6.1 Same job, different structural positions: writing and coding assistance

Several products serve the same broad job — help a user write or code — while occupying different layers. A standalone copy-generation tool (Jasper) sat largely on the surface (L7) and saw its differentiation compress when model providers shipped equivalent generation broadly — consistent with Law I. A grammar-and-style assistant (Grammarly) endured longer by owning distribution into browsers and editors (L4 + L7), until a larger owner of that same distribution layer (Microsoft, via the office suite) integrated a model directly. A coding assistant built into the editor (Cursor) fared differently again by accumulating workflow control, an indexing pipeline, an agent loop, and project memory (L4 + L6 + L8) — several reinforcing layers rather than one. Jobs-to-be-Done groups these products together; the layer map separates their exposure.

Corpus notes: Jasper / Grammarly / Copilot · Cursor

6.2 Data mispackaged as surface

Some communities accumulated valuable human knowledge but presented it as surface content (L7). When that content became training and retrieval input for model-layer systems (L2), an L1 data asset was captured economically by a layer owned by someone else, while the community that produced it captured little — a pattern visible in the post-2022 traffic decline of large developer-knowledge sites. The framework names the error: a value-bearing input was treated as content rather than protected as a scarce data layer.

Corpus note: Stack Overflow

6.3 Memory as the moat in enterprise agents

Two enterprise customer-service agents can look identical in a demonstration while differing structurally. One (Sierra) was architected from inception around behavioral data (L1), operating playbooks (L5), and per-customer compounding memory (L8) — the full triangle, where every resolution updates a customer-specific model. The other bolts an execution layer onto an existing data platform with no compounding loop. The first compounds; the second performs. The difference determines whether each use makes the next use more defensible.

Corpus note: Sierra vs. Agentforce

6.4 Single-layer fortresses

Not every durable position requires a triangle. A supplier of unusually scarce compute infrastructure (NVIDIA) defends an L0 position. A provider of legally cleared, licensed data (Adobe Firefly) defends an L1 + L3 position — the gate, not the generator, is the moat. A vertical data-plus-gate stack in a regulated domain (Bloomberg) defends L1 + L2 + L3. These cases discipline the framework against over-claiming the triangle: depth at one scarce layer, often far below the visible surface, is a second valid route to durability.

Corpus notes: Adobe Firefly · BloombergGPT

Table 2 collects these readings. The layer assignments are the author's structural judgment, not measurements; the same product could be placed slightly differently by another analyst (Section 7.2).

Company	Category	Layer position	Absorption risk	Framework reading
Jasper	AI writing	Surface only (L7)	High	Surface exposure; absorbed when generation went free
Grammarly	Writing assistant	Distribution + surface (L4 + L7)	Medium	Protected by distribution until a bigger L4 owner integrated
Cursor	Coding	Access + orchestration + memory (L4/L6/L8)	Lower	Reinforcing layers; the model is the commodity
Stack Overflow	Developer knowledge	Data mispackaged as content (L1→L7)	High	Value-bearing input captured by the ingesting layer
Sierra	Enterprise agents	Data + execution + memory (L1/L5/L8)	Lower	Full Defensible Triangle; compounds with use
NVIDIA	AI compute	Infrastructure (L0)	Low–Med	Single-layer fortress
Adobe Firefly	AI images	Licensed data + gate (L1/L3)	Med–Low	Gatekeeping moat, not the generator

Table 2. Illustrative archetypes. Layer assignments are structural judgment; trajectories are consistent with public reporting, not audited.

7. Limits and research agenda

A useful framework must name where it is weak. The Supply Chain of Intelligence is not a substitute for evidence, product taste, market timing, capital access, or execution quality.

7.1 Taxonomy, not causal proof

The framework classifies positions and proposes mechanisms; it does not prove that layer position causes survival, valuation, or margin. Because it is descriptive, no single case can falsify it the way a prediction could — a feature for a vocabulary and a liability for anyone tempted to treat it as a predictive model. It is not one.

7.2 Fuzzy boundaries

Layer assignment is partly judgment-based: a capability may sit between execution and orchestration; a data asset may become memory once connected to outcomes; a distribution gate may function as both access and gatekeeping. The framework reduces ambiguity relative to having no map, but any numeric score built on it should be treated as structured judgment, not measurement.

7.3 Moving bottlenecks

The layers are more stable than the scarcity of any particular layer. Synthetic data may relieve some data bottlenecks; model routing may commoditize parts of orchestration; regulation may create or erase gatekeeping positions; memory infrastructure may become easier to buy. A map drawn in 2026 should be redrawn as bottlenecks move.

7.4 Timing, capital, and execution

A company can occupy a strong position and still fail; another can occupy a weak position and survive on distribution, speed, capital, or execution. The framework locates *where* durable value can sit; it is largely silent on *when* and on *who executes well* — both of which often dominate outcomes.

7.5 Empirical agenda

The next version should become testable. One path is to code a population of AI-native products by primary layer, adjacent-layer ownership, data rights, trust gates, workflow depth, memory depth, and distribution control; outcomes could include retention, margin, acquisition, feature absorption, enterprise adoption, or valuation change, with controls for funding, distribution, team quality, incumbent response, and timing. The framework would then move from vocabulary to a set of hypotheses about product durability.

8. Conclusion

The Supply Chain of Intelligence proposes a simple reframing: an AI product is a position in a layered chain. The visible feature is rarely the whole product. Durable value tends to sit where the chain is scarce, trusted, embedded, or compounding. The four laws and the Defensible Triangle are devices for making those positions easier to see and discuss.

The framework should be judged by whether it improves decisions. Does it help a product leader see that a roadmap is overinvested in surface and underinvested in memory? Does it help a founder identify which layer must be owned before the current one commoditizes? Does it help an investor distinguish a thin interface from a workflow with accumulating advantage? If so, it earns practical value before it becomes an empirical theory. If not, the agenda in Section 7.5 shows where it must be sharpened.

Notes on method and attribution

This is a conceptual working paper. It reports no original dataset and conducts no statistical test. The framework, the four laws, and the Defensible Triangle are the author's synthesis. The public set of related diagrams and case notes is maintained at supplychainofai.com and may be cited or extended with attribution. Any future version making claims about specific companies, valuations, or performance trajectories should cite primary sources or clearly label the claims as illustrative.

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Appendix A. The fifty sublayers

Each layer decomposes into five sublayers. The framework marks sublayers that tend to be *structurally defensible* — those hardest for an adjacent layer to absorb — though, per Section 7.2, the boundaries are judgment-based.

Layer	Five sublayers
L8 Memory	Session & short-term memory · User & entity profiles · Aggregated network learning · Institutional knowledge · Learned world models
L7 Surface	Conversational · Visual interfaces & media · Embedded & copilot · Transaction surface · Async & ambient surfaces
L6 Orchestration	Agent loops · Human-in-the-loop · Role routing & task decomposition · Context & state management · Runtime assurance & learning loops
L5 Execution	Domain execution · Decision frameworks & reasoning scaffolds · Retrieval-augmented workflows · Operating playbooks · Interactional skills
L4 Access	API & integration layer · Agent interface protocols · Access governance · Real-time interaction infrastructure · Agent identity & provenance
L3 Gatekeeping	Compliance gates · Quality gates · Safety & security · Editorial gates · Distribution gates
L2 Models	Foundation models · Specialized models · Embedding & retrieval · Model routing & composition · Reasoning models
L1 Data	Public & open data · Proprietary data · Behavioral data · Outcome data · Synthetic data
L0 Infrastructure	Silicon · Data centers · Interconnect fabric · Compute access / cloud · Edge & on-device compute
L−1 Resources	Energy · Thermal & water management · Fabrication & foundry · Critical materials · Skilled trades & human capital

Table A1. Fifty sublayers across the eleven layers. Full descriptions are maintained at supplychainofai.com/framework.